Reg. No. | | | | | | | | | | |

[22MADIP02]

FOURTH SEMESTER B.Tech EXAMINATION, MAY–2024

FOUNDATION MATHEMATICS-II

Time:3 Hours **Maximum Marks:50**

SL. NO	QUESTIONS	A	B	C	D	ANSWER
1.	$\int_2^4\int_0^1 dxdy =$	2	4	8	None of these	
2.	$\int_2^3\int_{-1}^1 xdxdy =$	2	5	6	0	
3.	$\int_2^3\int_1^2 ydxdy =$	$\frac{7}{2}$	$\frac{3}{2}$	$\frac{5}{2}$	$\frac{9}{2}$	
4.	$\int_{-2}^3\int_0^1 xydxdy =$	$\frac{9}{2}$	$\frac{5}{4}$	$\frac{7}{2}$	None of these	
5.	$\int_4^5\int_2^3 x^2ydxdy =$	$\frac{57}{2}$	$\frac{59}{2}$	$\frac{9}{2}$	0	
6.	$\int_1^{-1}\int_{-2}^2 xy^2dxdy =$	0	3	4	14	
7.	$\int_3^{-4}\int_1^2 (x+y)dxdy =$	14	0	4	–14	
8.	$\int_1^{-1}\int_0^{\frac{\pi}{2}} \cos xdxdy =$	0	3	–2	1	
9.	$\int_1^4\int_0^{\infty} e^{-x}dxdy =$	1	2	4	3	
10.	$\int_{-2}^2\int_0^{-\infty} ye^xdxdy =$	2	0	3	6	

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11.	The Laplace Transform of $f(t)$ is	$\int_0^\infty e^{-st} f(t)dt$	$\int_0^\infty e^{-t} f(t)dt$	$\int_0^{2a} e^{-st} f(t)dt$	None of these	
12.	The value of $L[1]$ is	$\frac{4}{s}$	$\frac{3}{s}$	$\frac{2}{s}$	$\frac{1}{s}$	
13.	$L(e^{at})$ is equal to	$\frac{1}{s-a}$	$\frac{1}{s+a}$	$\frac{2}{s-a}$	$\frac{2}{s+a}$	
14.	$L(e^{-at})$ is equal to	$\frac{1}{s-a}$	$\frac{1}{s+a}$	$\frac{2}{s-a}$	$\frac{2}{s+a}$	
15.	$L(\sin at)$ is equal to	$\frac{a}{s^2-a^2}$	$\frac{a}{s^2+a^2}$	$\frac{s}{s^2-a^2}$	$\frac{s}{s^2+a^2}$	
16.	$L(\cos at)$ is equal to	$\frac{a}{s^2-a^2}$	$\frac{a}{s^2+a^2}$	$\frac{s}{s^2+a^2}$	$\frac{s}{s^2-a^2}$	
17.	$L(\sinh at)$ is	$\frac{1}{s^2+a^2}$	$\frac{a}{s^2-a^2}$	$\frac{s}{s^2+a^2}$	None of these	
18.	$L(\cosh at)$ is	$\frac{1}{s^2+a^2}$	$\frac{1}{s^2-a^2}$	$\frac{s}{s^2+a^2}$	$\frac{s}{s^2-a^2}$	
19.	$L(t^n)=$, Where n is non integer	$\frac{\Gamma(n)}{s^{n+1}}$	$\frac{\Gamma(n-1)}{s^{n+1}}$	$\frac{\Gamma(n+1)}{s^{n+1}}$	None of these	
20.	$L(e^{3t})$ is equal to	$\frac{1}{s-3}$	$\frac{1}{s+3}$	$\frac{2}{s-3}$	$\frac{1}{s-a}$	
21.	A coin is tossed three times in succession, the number of elements in the sample space	6	8	3	9	
22.	The probability of getting exactly one head in tossing a pair of coins is	2	0.5	1	0.333	
23.	If a card is drawn from a pack of cards, the probability of getting either a king or queen is	0.1538	0.1427	0.1316	0.0769	
24.	Mutually Exclusive events _____	Contain all sample points	Contain all common sample points	Does not contain any sample point	Does not contain any common sample point	
25.	The total number of events of throwing 10 coins simultaneously is.	1024	512	100	10	

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26.	The chance of throwing 5 with an ordinary die is	0.22	0.5	0.16	none of these	
27.	A coin is tossed twice. The probability of getting both heads is	0.25	0.333	0.5	none of these	
28.	The probability of a non-leap year having 53 Mondays is	0.14	0.22	0.5	none of these	
29.	A bag contains 5 black balls, 4 white balls and 3 red balls. If a ball is selected randomise, the probability that it is a black or red ball is	0.33	0.66	0.5	none of these	
30.	Which of the following matrices is in row reduced echelon form?	$\begin{bmatrix} 0 & 1 & 9 \\ 0 & 0 & 1 \\ 2 & 0 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & 3 & 0 & 7 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} 1 & 3 & 6 & 5 \\ 2 & 6 & 12 & 10 \\ 0 & 0 & 0 & 0 \end{bmatrix}$	None of these	
31.	The sum of the eigen values of a matrix s the sum of the elements of the	Principal diagonal	Determinant	Vectors	None of these	
32.	The determinant of $A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 3 \end{bmatrix}$ is....	-12	6	-28	None of these	
33.	Number of non-zero rows of a matrix is known as.....of a Matrix.	Diagonal	Normal	Rank	None of these	
34.	If A is a square matrix and $A^2 = I$, then A =	A = I	A = -I	$A \neq I$	None of these	
35.	Which of the following is not an elementary transformation?	Adding two rows	Squaring all the elements of the matrix	Adding two columns	Multiplying a row by a non-zero number	
36.	Rank of a matrix $A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & -1 & 4 \\ 1 & -11 & 14 \end{bmatrix}$	0	1	2	3	
37.	A set of linear equations is represented by the matrix equation Ax = b. The necessary condition for the existence of a solution for this system is.	A must be invertible	b must be linearly independent on the columns of A	b must be linearly dependent of the columns of A	None of these	

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38.	Consider the following two statements: i. The maximum number of linearly independent column vectors of a matrix A is called the rank of A. ii. If A is an n x n square matrix, it will be non-singular is rank A = n. With reference to the above statements, which of the following applies?	Both the statements are false	Both the statements are true	I is true but II is false	I is false but II is true	
39.	In the matrix equation Px = q. which of the following is a necessary condition for the existence of at least one solution for the unknown vector x?	Augmented matrix [Pq] must have the same rank as matrix P	Vector q must have only non-zero elements	Matrix P must be singular	None of these	
40.	Let V be a vector space, and let W be a subset of V. What does it mean when we say that W is closed under addition?	Whenever x & y are in W, then x+y is in V.	Whenever x and y are in V, then x+y is in W.	Whenever x And y are in W, then x+y is in W.	None of these.	
41.	Let V be a vector space, and let W be a subset of V. What does it mean when we say that W is closed under scalar multiplication?	Whenever x is in W and c is a scalar, then cx is in W.	Whenever x is in V and c is a scalar, then cx is in W.	Whenever x is in W and c is a scalar, then cx is in V.	None of these.	
42.	Let V be a vector space, and let S be a subset of V. What does it mean when we say that S is linearly independent?	S is a basis	only way to write 0 as a linear combination of elements of S is the zero combination	S is a basis and only way to write 0 as a linear combination of elements of S is the zero combination	None of these.	
43.	Let V be a vector space, and let S be a subset of V. What does it mean when we say that S is linearly dependent?	The span of S has smaller dimension than the dimension of V	At least two of the elements of S are the same	There is a way to write 0 as a linear combination of elements of S other than the zero combination.	None of these.	

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44.	Let V be a vector space, and let S be a subset of V. What does it mean when we say that S spans V?	Every vector in V can be expressed as a linear combination of vectors in S.	S is a basis for V	Every vector in V has exactly one representation as a linear combination of vectors in S.	None of these.	
45.	Let V be a five-dimensional vector space, and let S be a subset of V which spans V. Then S	Must be a basis for V	Must consist of at least five elements	Must have exactly five elements	None of these.	
46.	Let V be a five-dimensional vector space, and let S be a subset of V which is a basis for V. Then S	Must be linearly Dependent and independent	Must be linearly dependent.	Must be linearly independent.	None of these.	
47.	Let V be a five-dimensional vector space, and let S be a subset of V which is linearly independent. Then S	Must have at most five elements.	Must have at least five elements.	both A and B	None of these.	
48.	Let V be a five-dimensional vector space and let S be a subset of V which is a basis for V. Then S	Must have exactly five elements	Must have less than five elements	Can have any number of elements (except zero).	Must be linearly dependent.	
49.	Let V be a five-dimensional vector space and let S be a subset of V consisting of five vectors. Then S	Must be linearly independent, but may or may not span V	Must be linearly dependent, and must span V	Must be a basis of V	Can span V, but only if it is linearly independent, and vice versa	
50.	Let u, v, w be three non-zero vectors which are linearly independent, the	u is linear combination of v and w	v is linear combination of u and w	w is linear combination of u and v	None of these	